

Riemann Sums

Complete the following steps for $f(x) = 24 - 2x$, interval $[0, 6]$, and $n = 3$.

1. Sketch the graph of the function on the given interval and illustrate the *right Riemann sums* using n subintervals.
2. Calculate the *right Riemann sums* for the given interval and number of subintervals, n .

Understanding Integration

Graph an integrand and use geometry to evaluate the integral.

3. $\int_1^3 2x \, dx$ (use $\frac{1}{2}bh = \text{area of a triangle}$)

The figure shows the area of regions bounded by the graph of $f(x)$. Evaluate the following integrals.

4. $\int_0^8 f(x) \, dx$

5. $\int_0^5 f(x) \, dx - \int_8^0 f(x) \, dx$

Evaluating indefinite/definite integrals.

6. $f(x) = 9x^8$

a. $\int 9x^8 \, dx$

b. $\int_0^1 9x^8 \, dx$

7. $f(x) = x^2 - 8x + 1$

a. $\int (x^2 - 8x + 1) \, dx$

8. $f(x) = \cos 3x$

a. $\int \cos 3x \, dx$

b. $\int_{-\pi/3}^{\pi/3} \cos 3x \, dx$

9. $f(x) = \frac{x - \sqrt{x}}{x^3}$

a. $\int \frac{x - \sqrt{x}}{x^3} \, dx$

10. $f(x) = x^2(3x^3 + 1)^4$

a. $\int x^2(3x^3 + 1)^4 \, dx$

b. $\int_1^2 x^2(3x^3 + 1)^4 \, dx$

11. $v(\theta) = \cos \theta \sin^2 \theta$

a. $\int \cos \theta \sin^2 \theta \, d\theta$

12. $g(t) = \frac{t}{\sqrt{4 - t^2}}$

a. $\int \frac{t}{\sqrt{4 - t^2}} \, dt$

b. $\int_2^4 \frac{t}{\sqrt{4 - t^2}} \, dt$

Average value of a function over an interval.

13. A hiking trail has an elevation given by $h(x) = 40x^3 - 300x^2 + 200x + 4000$ where h measures feet above sea level and x represents miles along the trail with $0 \leq x \leq 5$. What is the average elevation along the trail?

Initial value problems

Solve the initial position problem .

14. $\frac{dS}{dt} = (t + 9)^{1/2}$ with $S(0) = 20$. Find $S(t)$.

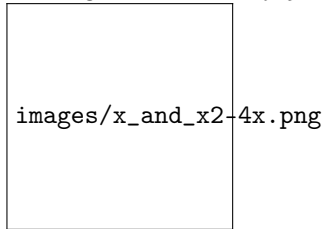
Solve the problem.

15. A projectile is launched vertically from the ground at $t = 0$ where t is seconds, and its velocity in flight (in m/s) is given by $v(t) = 20 - 10t$.
- a. Find the position at time t for $0 \leq t \leq 4$.
 - b. Find the total distance traveled from $t = 0$ to $t = 4$.

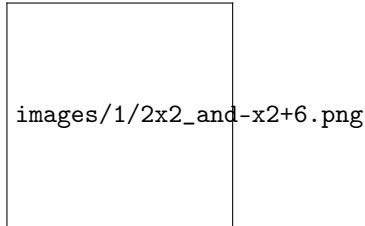
Area between graphs

Use calculus to find the area of the region described.

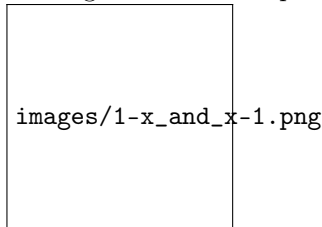
16. The region bounded by $y = x$ and $y = x^2 - 4x$.



17. The region bounded by $y = \frac{1}{2}x^2$ and $y = -x^2 + 6$.



18. The region in the first quadrant bounded by $-x + 1$ and $x - 1$ from $0 \leq x \leq 2$.

**Volume of Solids**

Let R be the region bounded by the following curves. Use the disk method to find the volume of the solid generated when R is revolved about the x -axis.

19. $y = \sqrt{x}$, $x = 0$, and $x = 9$.
 20. $y = \sqrt{\sin 4x}$, $y = 0$, $0 \leq x \leq \frac{\pi}{4}$.

Length of the curve

Find the length of the curve.

21. $y = 2x^{3/2}$ from $x = 0$ to $x = \frac{5}{4}$.
 22. $y = \left(4 - x^{2/3}\right)^{3/2}$ from $x = 0$ to $x = 8$.

Derivatives and Integrals of $\ln x$ and e^x .

23. $\frac{d}{dx}(\ln x + 1)$

24. $\frac{d}{dx} 3e^x \ln x$

25. $\frac{d}{dx}(\ln(\cos x) - 2x)$

26. $\frac{d}{dx} e^{-2x+x^2}$

27. $\frac{d}{dt} \frac{e^t}{e^t + 1}$

28. $\int \left(3e^{-3x} + \frac{1}{x} \right) dx$

Integration by parts

29. $\int x e^{-4x} dx$

30. $\int x \cos 4x dx$

31. $\int \frac{4 \ln x}{x^5} dx$

32. $\int_0^{\ln 1} 3x e^x dx$

Integrate functions with even/odd powers of $\sin x$ and $\cos x$.

33. $\int \ln 3x dx$

34. $\int \sin^2 3x dx$

35. $\int \cos^3 x dx$

36. $\int \sin^2 x \cos^2 x dx$

37. $\int \cos^3 x \sin x dx$

Integrate functions use Trigonometric substitution.

38. $\int \frac{dx}{\sqrt{4-x^2}}$

39. $\int_0^4 \sqrt{16-x^2} dx$

40. $\int \sqrt{16-x^2} dx$

Method of Partial Fractions.

41. $\int \frac{8x+22}{x^2+4x-5} dx$

$$\int \frac{8x+22}{x^2+4x-5} dx = \int \frac{8x+22}{(x-1)(x+5)} dx \text{ and } \frac{8x+22}{(x-1)(x+5)} = \frac{A}{x-1} + \frac{B}{x+5}.$$

$$\Rightarrow A(x+5) + B(x-1) = 8x+22$$

$$\Rightarrow Ax + 5A + Bx - B = 8x + 22, \text{ clearly then } Ax + Bx = 8x \text{ and } 5A - B = 22.$$

Solving the first one for A , $A = 8 - B$ and plugging this into the second gives, $40 - 5B - B = 22 \Rightarrow B = 3 \Rightarrow A = 5$.

Now we have, $\frac{8x+22}{(x-1)(x+5)} = \frac{5}{x-1} + \frac{3}{x+5}$, and we find $\int \frac{5}{x-1} + \frac{3}{x+5} dx = 5 \ln |x-1| + 3 \ln |x+5|$.

42. $\int \frac{6}{x^2+2x-8} dx$